

①

* Card and Krueger's DID.

$$\begin{aligned}
 DID &= (\bar{Y}_{\text{After, NJ}} - \bar{Y}_{\text{Before, NJ}}) - (\bar{Y}_{\text{After, PA}} - \bar{Y}_{\text{Before, PA}}) \\
 &= (20.90 - 20.43) - (21.10 - 23.38) \\
 &= \boxed{2.75} \leftarrow \text{policy effect.}
 \end{aligned}$$

(in fast-food restaurants)
 Treatment state
 control state.
 After the policy change
 Before the policy change

Note that we can also obtain the policy effect by:

$$\begin{aligned}
 &(\bar{Y}_{\text{After, NJ}} - \bar{Y}_{\text{After, PA}}) - (\bar{Y}_{\text{Before, NJ}} - \bar{Y}_{\text{Before, PA}}) \\
 &= (20.90 - 21.10) - (20.43 - 23.38) = 2.75.
 \end{aligned}$$

* Define an interaction regression for DID.

Let us define two dummy variables D_1 and D_2 .

$$D_1 = \begin{cases} 1 & \text{Treatment Group/State (NJ)} \\ 0 & \text{Control Group/State (PA).} \end{cases}$$

$$D_2 = \begin{cases} 1 & \text{After the policy change} \\ 0 & \text{Before the policy change.} \end{cases}$$

Then, define the interaction regression model:

$$Y = \beta_0 + \beta_1 D_1 + \beta_2 D_2 + \beta_3 D_1 \times D_2 + u.$$

(2)

* Now, let us verify that β_3 is the DID parameter.

$$\{0,0\}: E[Y | \underbrace{D_1=0}_{\text{PA}}, \underbrace{D_2=0}_{\text{Before}}] = \cancel{\beta_0} \text{ (1)}$$

$$\{1,0\}: E[Y | \underbrace{D_1=1}_{\text{NJ}}, \underbrace{D_2=0}_{\text{Before}}] = \cancel{\beta_0} + \cancel{\beta_1} \text{ (2)}$$

$$\{0,1\}: E[Y | \underbrace{D_1=0}_{\text{PA}}, \underbrace{D_2=1}_{\text{After}}] = \cancel{\beta_0} + \cancel{\beta_2} \text{ (3)}$$

$$\{1,1\}: E[Y | \underbrace{D_1=1}_{\text{NJ}}, \underbrace{D_2=1}_{\text{After}}] = \cancel{\beta_0} + \cancel{\beta_1} + \cancel{\beta_2} + \beta_3 \text{ (4)}$$

$$\text{Therefore, } \beta_3 = [(4) - (3)] - [(2) - (1)]$$

$$= [\{ \text{NJ, After} \} - \{ \text{PA, After} \}] - [\{ \text{NJ, Before} \} - \{ \text{PA, Before} \}]$$

equivalent.

which is equivalent to:

$$[\{ \text{NJ, After} \} - \{ \text{NJ, Before} \}] - [\{ \text{PA, After} \} - \{ \text{PA, Before} \}]$$

$$\text{as } [(4) - (3)] - [(2) - (1)] = [(4) - (2)] - [(3) - (1)]$$

This verifies that β_3 is the DID parameter,

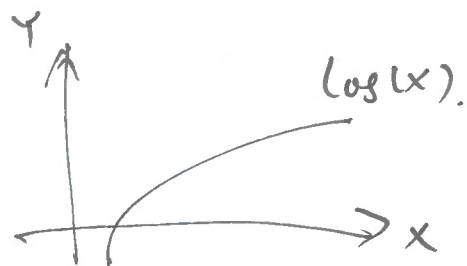
so in practice, we can run an OLS regression,

and $\hat{\beta}_3$ is our DID estimate for the policy effect, from the OLS regression.

③ * Linear-Log model.

$$Y_i = \beta_0 + \beta_1 \log(X_i) + u_i$$

$\uparrow$ $\uparrow$
Linear Log.



$$\beta_1 \approx \frac{\text{change in } Y}{\text{percentage change in } X} = \frac{\Delta Y}{\Delta x / x}$$

$$\Rightarrow \Delta Y \approx \beta_1 \cdot \frac{\Delta x}{x}$$

e.g., suppose $\beta_1 = 2$, and $\frac{\Delta x}{x} = 10\%$.

$$\Rightarrow \Delta Y \approx 2 \cdot 10\% = 0.2 \text{ (unit of } Y \text{)}.$$

Derivation (no need to reite)

$$\left. \begin{aligned} Y &= \beta_0 + \beta_1 \cdot \log(x) \\ Y + \Delta Y &= \beta_0 + \beta_1 \cdot \log(x + \Delta x) \end{aligned} \right\}$$

$$\Rightarrow \Delta Y = \beta_1 [\log(x + \Delta x) - \log(x)]$$

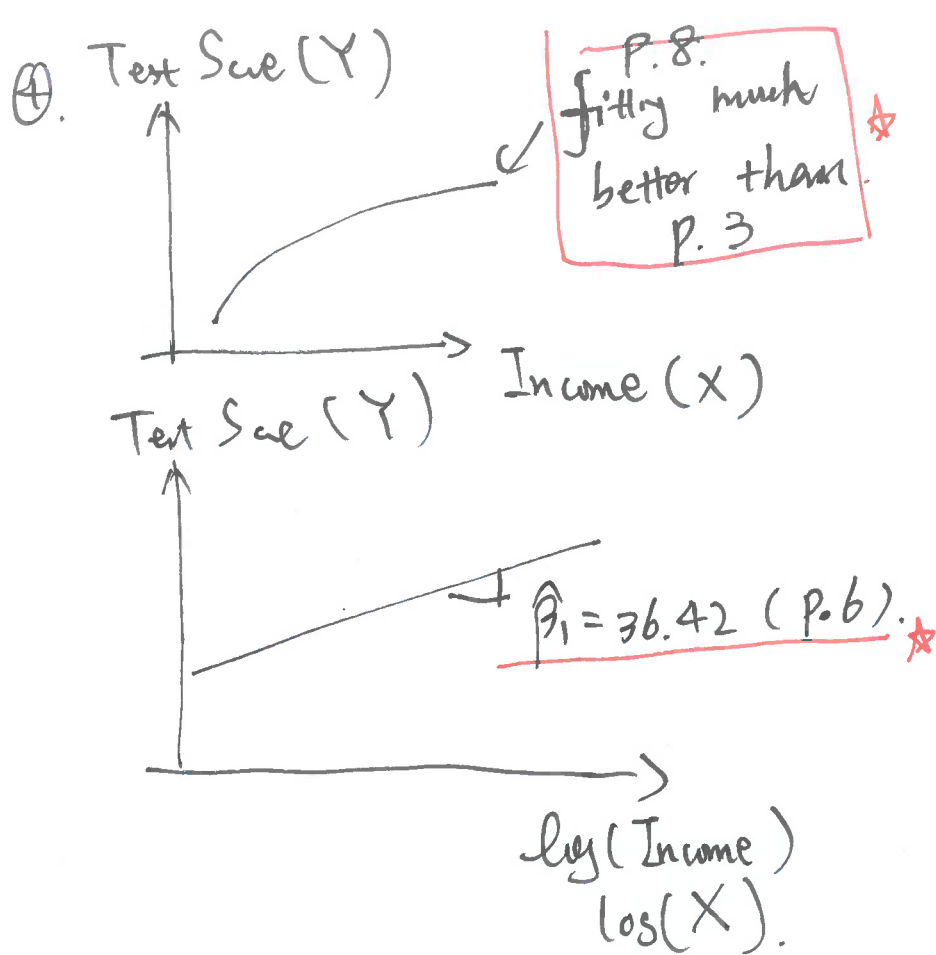
$\approx \frac{\Delta x}{x} \quad (d \log(x) = \frac{1}{x} dx)$

* Example Test Score & Income.

$$\hat{\beta}_1 = 36.42$$

$$\hat{\Delta Y} \approx \hat{\beta}_1 \cdot \frac{\Delta x}{x} = 36.42 \times 1\% = 0.3642 \text{ (point)}$$

unit of test score
↓



Example: EMH.

$$\hat{\beta}_3 = -4.74. \quad \beta_3 \approx \frac{\Delta Y}{\Delta X_3 / X_3}$$

$$\hat{\Delta Y} \approx \hat{\beta}_3 \cdot \frac{\Delta X_3}{X_3} = -4.74 \cdot 1\% = -0.0474 (\$)$$

$$\hat{\beta}_4 = 7.24. \quad \beta_4 \approx \frac{\Delta Y}{\Delta X_4 / X_4}$$

$$\hat{\Delta Y} \approx \hat{\beta}_4 \cdot \frac{\Delta X_4}{X_4} = 7.24 \cdot 1\% = 0.0724 (\$)$$

$$F = \frac{R_u^2 / 4}{(1 - R_u^2) / (142 - 5)} = 1.1688 < 2.37 \quad \begin{matrix} 5\% \\ \text{critical value} \\ \text{for } F_{4,137} \end{matrix}$$

can't rej. H_0 that $\beta_1 = \beta_2 = \beta_3 = \beta_4 = 0$
at the 5% significance level.

suppose the unit
of Y is dollar